F79SP Session 16: Branching processes

Branching processes, or Galton-Watson processes are a class of interesting stochastic processes which model population growth.

1 Discrete-time branching processes

A discrete-time branching process $\{X_n\}_{n\geq 0}$ is modeled as follows:

- At each integer time n, we have a number X_n of individuals, each individual will gives birth to a random number of offspring at time $n+1$, independently from one another. The parent dies immediately upon giving birth.
- Here, the "branching time" is fixed to be 1 for all individuals.
- Denote the offspring of ith individual in n-th generation as the r.v. $\xi_i^{(n)}$ (which takes non-negative integers as value).
- Then, $X_{n+1} = \xi_1^{(n)} + \cdots + \xi_{X_n}^{(n)} = \sum_{i=1}^{X_n} \xi_i^{(n)}$.

The branching process defined above has **Markov property.**

$$\begin{aligned}
&\mathbf{P}(X_{n+1} = j \mid X_n = i, X_{n-1} = i_{n-1}, \ldots, X_0 = i_0) \\
&= \mathbf{P}(\sum_{r=1}^{X_n} \xi_r^{(n)} = j \mid X_n = i, X_{n-1} = i_{n-1}, \ldots, X_0 = i_0) \\
&= \mathbf{P}(\sum_{r=1}^{i} \xi_r^{(n)} = j \mid X_n = i, X_{n-1} = i_{n-1}, \ldots, X_0 = i_0) \\
&= \mathbf{P}(\sum_{r=1}^{i} \xi_r^{(n)} = j) =: p_{ij}
\end{aligned}$$

- The probability is a function of i and j. It does not depend on $i_{n-1}, \ldots$ and n.
- Hence, $\{X_n\}_{n\geq 0}$ is a Markov chain.

2 Some trivial examples

Throughout, we denote:

$$p_j = \mathbf{P}(\xi_i^{(n)} = j)$$

be the probability that an individual will bear j offspring, where $j = 0, 1, 2, \ldots$.

We thus have:

$$\sum_{j=0}^{\infty} p_j = 1.$$

Example 2.1. *Consider a DT branching process with $p_1 = \boldsymbol{P}(\xi = 1) = 1$. Then:*

Example 2.2. *Consider a DT branching process with $p_2 = \boldsymbol{P}(\xi = 2) = 1$. Then:*

Example 2.3. *Suppose that $p_1 = p$, $p_0 = 1 - p$ and $p_k = 0$, for $k \geq 2$.*

- *Here, each individual either gives birth to one child (w.p. p) or has no children (w.p. $(1 - p)$).*
- *Given $X_n = i$, then X_{n+1} is the number of successful births amongst i individuals, which follows $B(i, p)$.*
- *The population cannot grow.*
- *The population will eventually reach 0, that is, extinct.*
- *We say 0 is an absorbing state.*

3 Interesting questions

Given a branching process, the following questions are often of interest:

1. What is the expected time to reach a certain population?
2. What is the expected time to extinction?
3. What is the probability of extinction?

3.0.1 The expected time to extinction

Reconsider the case when $p_1 = p$, $p_0 = 1 - p$ and $p_k = 0$, for $k \geq 2$. Extinction is certain here.

- Let $X_0 = i$.
- Let τ_l = time to die for the lth branch.

3.0.2 The expected time to reach a certain population

This is similar to the expected hitting time in the context of Markov chains, which can be calculated using the conditioning on first step technique. We consider an example as follows.

Example 3.1. *Consider the Galton-Watson process with $X_0 = 1$, $p_1 = \frac{1}{3}$, $p_2 = \frac{1}{4}$ and $p_3 = \frac{5}{12}$. Let τ_i be the expected time to hit $S = \{3, 4, 5, \dots\}$, given that $X_0 = i$.*

1. *Determine τ_1 and τ_2.*